DentAssist: An Educational Prototype Combining Computer Vision and Agent-Based Modelling for Panoramic Dental X-Ray Workflows

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***Abstract*—Panoramic dental X-rays are commonly used to provide information on the state of the teeth and jaw. However, they require significant time to interpret, and determining the best way to route cases within a clinic can be challenging. Several published studies report only predictive scores. The dataset we selected has significant overlap between its training and test sets because they share source X-rays, meaning the reported test scores are not reliable. We present DentAssist, an educational prototype that combines computer vision and agent-based modelling (ABM). After image enhancement, the images are passed through the Canny edge detection algorithm. A YOLO image segmentation model is then used to predict masks, labels and confidence scores for selected dental findings. These outputs serve as inputs to a Python simulation with patient, screening and dentist agents, allowing manual and AI-assisted workflows to be compared under the same arrival rates and staffing levels. Our audit found that 1,509 of the 1,573 X-rays in the test set had also been used in the training set (13,932 images in total, 31 classes). We are currently creating a leakage-free split, retraining the model and running experiments. As this is a teaching prototype, no model or workflow results are presented. The code will be made publicly available.**

***Index Terms*—agent-based modelling, data leakage, dental radiography, edge detection, instance segmentation, panoramic X-ray, YOLO.**

I. INTRODUCTION

Panoramic radiography, or orthopantomography (OPG) [1], produces a single image of the whole mouth, including the teeth, jaws and supporting bone. Dentists use these images to spot common dental problems such as caries, missing or impacted teeth, and bone loss. Because one panoramic image can display many different findings, automatically locating them can help reviewers focus on the most relevant parts of the image.

Computer vision techniques can be used to boost contrast and find edges within the image [2], which helps to identify the teeth and bones. Instance segmentation produces a mask for every object together with a class label and a confidence score. YOLO-style models make fast predictions [3], [4], which makes them well suited to a lightweight interactive application.

In a clinical environment, cases arrive over time and enter a waiting queue to be reviewed by a small number of dentists. Agent-based modelling is a natural choice for simulating this process. Rather than describing the whole clinic with a single equation, ABM creates multiple agents (patients and reviewers) and applies simple rules to each one, so that the behaviour of the overall process emerges from their interactions [5].

Two gaps exist in this area. First, many dental X-ray studies and datasets report only detection and segmentation scores and do not explain what these scores mean for a clinical workflow [6]–[8]. Second, results depend heavily on how the dataset is split. Our audit of the chosen dataset Dental Disease Panoramic Detection Dataset — by Lokisilvres on Kaggle. (13,932 images and 136,891 annotated instances across 31 classes) found that its splits were biased: 1,509 (95.93%) of the 1,573 source radiographs in the test set also appeared in the training set, and 57.38% of the validation sources overlapped with the training set. Such leakage inflates reported scores [9], so a new split must be created at the level of source images.

DentAssist is designed to address these issues. Fig. 1 shows how image processing connects to the simulated clinic workflow. Within the CV module, candidate findings are identified for a number of high-confidence classes. The ABM module then uses the number of findings and their confidence scores to simulate cases. This is a simulation study: it explores different scenarios under stated assumptions but does not measure improvements in a real clinical setting.

[Insert Figure 1: DentAssist pipeline — CV module → ABM module]

Fig. 1. Overview of DentAssist: the computer vision module produces candidate findings and confidence scores, which drive case generation in the agent-based clinic simulation.

A. Contributions

The main contributions of this work are as follows:

- **Dataset audit and leakage-free split.** We audited a panoramic dataset of 13,932 images and are constructing a new train/validation/test split based on source images, with no overlap between subsets.
- **Computer vision pipeline.** Pre-processing, edge detection and YOLO segmentation have been applied to a number of classes and integrated into an interactive Streamlit application.

- **Agent-based clinic model.** We built an agent-based model in Python with patient, screening and reviewer agents to compare manual and AI-assisted workflows.

B. Scope and Limitations

The model can only predict classes present in its training data, and its predictions do not constitute a diagnosis. At this stage, the dataset has been audited and the system designed; training, evaluation and integration will be carried out in future work.

The remainder of this paper is organised as follows. Section II describes the dataset. Section III details the methodology. Section IV describes how the system will be evaluated. Section V concludes the paper and discusses its limitations.

II. DATASET

The project uses the Dental Disease Panoramic Detection Dataset from Kaggle, containing 13,932 images across 31 original classes. Ten dental categories were selected for model training and detection.

III. METHODOLOGY

OpenCV performs grayscale conversion, CLAHE, filtering and Canny edge detection, followed by YOLOv8 segmentation. ABM simulates patient screening, priority queues and dentist workflows.

IV. EVALUATION PLAN

The CV model is evaluated using precision, recall and mAP metrics. ABM compares manual and AI-assisted workflows using waiting time, queue length and dentist utilization.

V. CONCLUSION AND LIMITATIONS

DentAssist combines dental X-ray detection with clinic workflow simulation. Limitations include uneven detection accuracy, dataset dependency and the need for clinical validation.

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